

Newton School of Technology

Diagnostics with ICMP, and the Move to IPv6

Join the lecture online on your dashboard

We Built It. Now We Test It.

Newton School of Technology

Behind us

CIDR · Subnets · NAT · Routes

We can address hosts, carve networks, and move packets between them.

→

Today · two questions

- **How do we test and troubleshoot a network?** → ICMP, ping, traceroute.
- **What's the long-term fix for the address shortage?** → IPv6.

NAT was a patch for running out of addresses. What's the actual cure?

Newton School of Technology

ICMP — The Network's Messenger

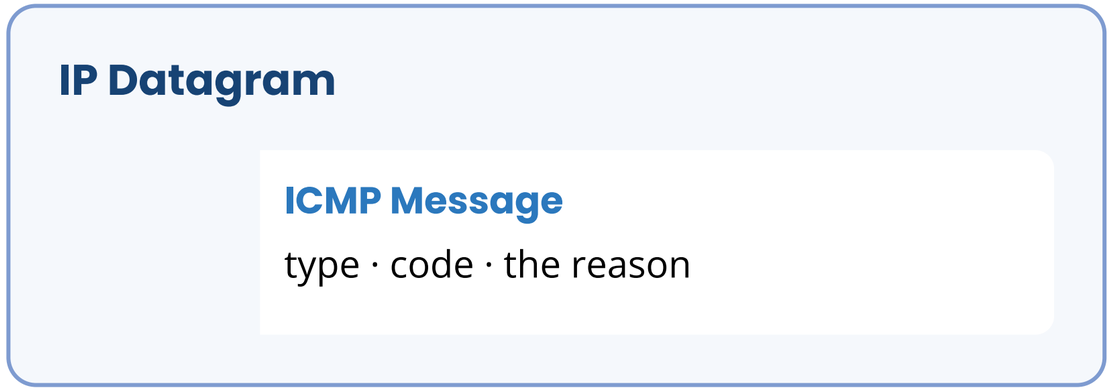
How routers and hosts tell each other what went wrong.

What Is ICMP?

Newton School of Technology

IP's control & error channel — not a data channel.

- Carries **control and error messages** for IP, never user data.
- It resides **inside** IP datagrams, yet it's a network-layer helper.
- A router that drops a packet can say **why** — that's the root of diagnostics.



ICMP travels in IP — but reports on IP

If a packet dies silently, you're blind. ICMP is how the network speaks up.

The Messages It Carries

Newton School of Technology

Echo Request / Reply

"Are you there?" → "Yes." The pair that powers **ping**.

Time Exceeded

TTL hit 0 — a router drops it and reports back. Powers **traceroute**.

Destination Unreachable

No route, host down, or port closed — delivery failed.

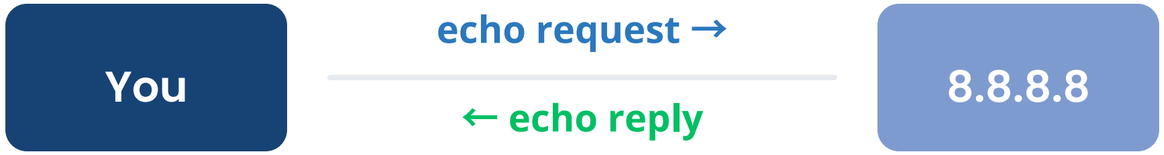
Redirect

"There's a better next hop" — sent by a router to a host.

ping — Are You There?

Newton School of Technology

Reachability, plus a stopwatch on the round trip.



- **Echo request** out → **echo reply** back.
- Confirms the host is **reachable**.
- Measures **round-trip time** — the same propagation delay from Lecture 2.

```
$ ping -c 3 8.8.8.8
64 bytes from 8.8.8.8: icmp_seq=1 time=12.4 ms
64 bytes from 8.8.8.8: icmp_seq=2 time=11.8 ms
64 bytes from 8.8.8.8: icmp_seq=3 time=12.1 ms
rtt min/avg/max = 11.8/12.1/12.4 ms
```

RTT is real distance-in-time: farther host, bigger number.

Newton School of Technology

Two Pings, Two Stories

The reply — or the silence — is the answer.

- A reply tells you: **reachable**, and how long the round trip took.
- Silence tells you something too — the packet went out and nothing came back.
- Your first question in any outage: **did anything answer?**

```
$ ping 8.8.8.8
64 bytes from 8.8.8.8: time=12.4 ms
64 bytes from 8.8.8.8: time=11.9 ms
3 packets sent, 3 received, 0% loss
```

```
$ ping 10.55.99.200
Request timed out.
Request timed out.
3 packets sent, 0 received, 100% loss
```

What does a fast reply prove? What does total silence prove?

Traceroute — The TTL Trick

Newton School of Technology

Send packets designed to die — and listen to who reports the death.



Each round reveals one more router. Rising TTL walks the whole path.

Newton School of Technology

Where Does the Path Break?

- **Rising RTTs** — each hop is farther away.
- *** * *** — no reply: filtered or dropped, not always broken.
- The **last responding hop** is your prime suspect.

```
$ traceroute example.edu
1 router.lan 1.2 ms
2 isp-edge 8.7 ms
3 core1.isp 14.3 ms
4 core2.isp 22.9 ms
5 peer.transit 41.6 ms
6 border.example 58.1 ms ← last reply
7 * * *
8 * * *
9 * * * (path stops here)
```

Newton School of Technology

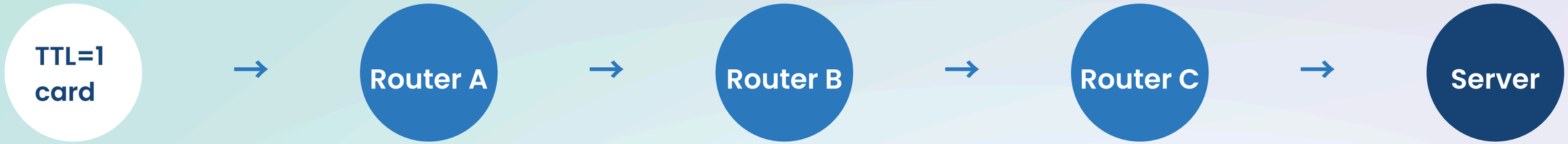
Replies through hop 6, then * * * to the end. Where do you start investigating?

At or just after hop 6 — the last hop that spoke.

Traceroute, Performed

Newton School of Technology

The class becomes the path. Watch it discover itself, one hop per round.



1 · Decrement

Each "router" subtracts 1 from the TTL card as it passes.

2 · Hit zero → shout

Whoever reaches 0 raises a hand: "Time Exceeded — I'm hop N."

3 · Re-send, +1

Send TTL=2, then 3. The reveal walks one hop further each time.

Then run the real traceroute — and match it to what the room just did.

Newton School of Technology

IPv6 — The Real Fix for Exhaustion

More addresses than we could ever assign.

Problem: 32 Bits Wasn't Enough

Newton School of Technology

IPv4 space

2^32

≈ 4.3 billion addresses

fewer than people on Earth

NAT — the patch



Many hosts share one address — clever, but it hides them from the outside.

NAT bought us time. It didn't give us more addresses.

128 Bits, Effectively Unlimited

Newton School of Technology



8 groups × 4 hex digits · 16 bits each = **128 bits**

total addresses $\approx 3.4 \times 10^{38}$

Enough for every device, sensor, and thing — with room to spare.

Read it, know why it's vast — don't hand-calculate the subnets.

Reading IPv6: The :: Shortcut

Newton School of Technology

Full

```
2001:0db8:0000:0000:0000:0000:0000:0001
```

1 · Drop leading zeros

```
2001:db8:0:0:0:0:0:1
```

One rule

:: can appear **only once** per address — otherwise the zero count is ambiguous.

Your turn: compress 2001:0db8:0000:0000:0000:0000:0000:0001.

What Actually Changes

Newton School of Technology

No NAT needed

End-to-end addressing returns — every host is directly reachable.

Simplified header

Fixed, streamlined format — faster for routers to process.

Autoconfiguration

Hosts can self-assign an address — no DHCP server required.

No broadcast

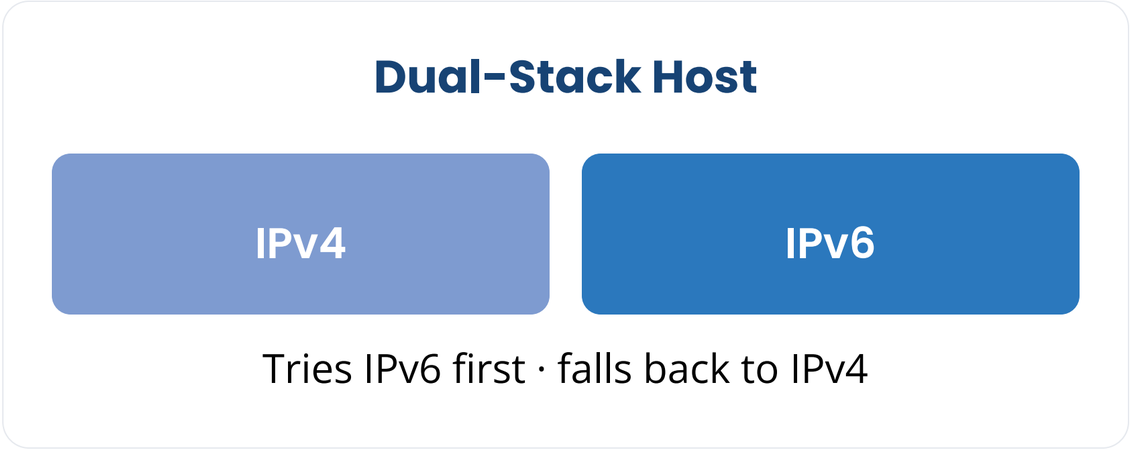
Uses **multicast** instead — talk to a group, not the whole wire.

IPv6 doesn't just add room — it simplifies the plumbing.

Living in a Mixed World

Newton School of Technology

- **Dual stack** — run v4 and v6 together; prefer v6, fall back to v4.
- Tunneling & translation bridge the gaps where needed.
- Adoption is slow: NAT "worked well enough," and the net can't flip overnight.



Cloud note

AWS VPCs take an IPv6 CIDR block alongside IPv4 — same subnetting thinking, far more room.

The future is here — it's just running quietly next to the past.

Key Takeaways

Newton School of Technology

1. ICMP is IP's control/error channel: **ping** tests reachability + RTT, **traceroute** maps the path with the TTL-expiry trick.

2. When troubleshooting, the **last responding hop** in a traceroute is your prime suspect.

3. IPv6 (128-bit) is the real cure for exhaustion — restoring end-to-end addressing — rolled out mostly via **dual stack**.

Please fill the feedback form.

Thank You